

Partisanship and Older Americans' Engagement with Dubious Political News

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Abstract Studies based on digital trace data show that older Americans visit and share dubious news sources far more often than younger cohorts, tendencies often attributed to lower levels of digital literacy. At the same time, survey experiments show that older Americans are no worse, if not better, at discerning between false and accurate news. If older Americans can identify misleading news content equally well, why are they still *more* likely to engage with it in observational settings? In this article, we combine survey measures and digital trace data for three nationally representative samples (N = 9,944) to argue that the existing literature overemphasizes the importance of factors like digital literacy relative to standard political variables such as political interest and partisanship, factors known to increase across the lifespan. Calcified partisanship in particular makes older Americans vulnerable to hyperpartisan news—which is highly slanted but not verified as explicitly false. High rates of engagement with this category of content, which has been examined in survey studies of older citizens less regularly in the literature, may partially explain the high rates of engagement with dubious news domains in behavioral trace data. In all, our findings have important implications for how we understand—and might intervene to reduce—high engagement among this cohort with dubious news.

Concerns about the public's susceptibility to untrustworthy, false, or otherwise dubious news¹ online have motivated an explosion of research focusing on

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1. We use the specific terms “hyperpartisan news” and “false news” when referring to our headline stimuli (see Methods). We define “false news” narrowly as content that has been fact-checked as false. We use the more inclusive word “dubious” to refer to observed news exposure

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vulnerable subgroups, especially older news consumers (Allen et al. 2020; Loos and Nijenhuis 2020). Compared to younger users, older Americans shared links to and visited such sites more often in 2016 and 2020 (Grinberg et al. 2019; Guess, Nagler, and Tucker 2019a; Guess, Nyhan, and Reifler 2020; Guess et al. 2021). Why are older news consumers especially vulnerable to dubious content? Brashier and Schacter (2020) discuss potential culprits that include cognitive declines, social changes, and digital illiteracy. Poor digital literacy among older news consumers, in particular, has attracted the attention of academics (Lee 2018; Moore and Hancock 2020, 2022), nonprofits (e.g., MediaWise for Seniors from Poynter), and the press (Span 2020).

Yet, the evidence supporting these skills-based explanations for age differences remains mixed. Lab studies, for instance, suggest that older adults are *no* more likely than younger counterparts to perceive false news to be accurate (Pehlivanoglu et al. 2022). Further, reanalysis (Brashier and Schacter 2020) of multiple survey studies in the United States (Allcott and Gentzkow 2017; Pennycook and Rand 2018) shows that the ability to distinguish legitimate and false headlines *increases* with age, a finding recently replicated cross-nationally (Arechar et al. 2023; Peren Arin, Mazrekaj, and Thum 2023) (see also exploratory analyses in Badrinathan 2020; Calvillo et al. 2020). In other words, the problem likely is *not* that older cohorts are simply less able to identify false news content.

There are multiple possible explanations for conflicting findings over age-related differences between these different types of studies, ranging from the narrowly methodological to the theoretical. These literatures tend to rely on different sampling strategies, and we are aware of no previous research that uses both approaches—digital trace data and survey ratings tasks to assess misinformation discernment—on a single representative sample. Some research even shows a disconnect between news content that is believed and content that is engaged with (read or shared) (Pennycook and Rand 2021; Sirlin et al. 2021), suggesting different antecedents for each.

More fundamentally, however, there are many variables that may correlate with both age and engagement with low-quality news content; it is not clear that digital literacy is the most likely culprit from among these factors. For instance, older Americans are also more politically engaged and consume more media overall (Glenn and Grimes 1968; Sears and Funk 1999; Hobbs 2019; Moretto et al. 2022). Most importantly, older Americans have more calcified partisan identities and report higher levels of affective polarization (Phillips 2022), factors that may affect what news older Americans engage

that is of questionable provenance (i.e., from low-quality news domains that we examine in our trace data) but not necessarily fact-checked as false. In our background and discussion sections, we sometimes refer to misinformation when making a general point drawn from the literature. We use “fake” news only in quotes when prior work employed the term.

with. Perhaps it is the concentration of these factors in older cohorts that is driving engagement?

In this article, we use three nationally representative surveys of Americans with nearly 10,000 respondents, linked directly to trace data of online behavior. These data allow us to examine real-world online behavior and performance on survey-based news discernment tasks *among the same set of respondents*. We find that older Americans are indeed more likely to engage with dubious news content; yet older Americans are no more likely to be deceived in news discernment tasks. Moreover, there is no evidence that digital literacy training is more effective in this cohort. Both results are inconsistent with the idea that the inability to correctly identify false news content is primarily driving engagement with dubious sources, although it may still play a role.

Instead, we argue that a key underlying factor in older Americans' disproportionate engagement with dubious news online is increasingly hardened partisanship (Phillips 2022). Older Americans in our sample exhibit significantly higher levels of affective polarization and congeniality bias toward stories consistent with their partisan leanings. In particular, this group is more likely to believe congenial hyperpartisan news—which is highly slanted but not rated as explicitly false. Since many low-quality web domains examined in trace data studies mix false and hyperpartisan news content, our results suggest that it may be a preference for content that reinforces partisan identities rather than illiteracy that is responsible for the high levels of engagement we observe in older Americans. Our findings call into question existing common wisdom. Our findings suggest that researchers should take more seriously the role of hyperpartisan—rather than explicitly false—news content in digital trace data (Ross, Rand, and Pennycook 2021) when considering the nature of news diets and sources of misperceptions. Further, interventions drawn from research on affective polarization (Voelkel et al. 2023) may be important in addressing the disproportionate engagement with dubious content in older cohorts.

Vulnerability to Misinformation across the Lifespan

It is first important to justify a focus on older adults' online behavior, given their disproportionate interest in traditional media (Muise et al. 2022). Although older adults watch far more television news than younger audiences do (Muise et al. 2022), they also engage in significant online news media use. In fact, recent consumption data shows that older adults attend to even more online news than younger audiences do (Allen et al. 2020). Importantly, most news, and partisan news in particular, that is shared online comes from older adults (Moretto et al. 2022). As such, their digital behavior is important to understand.

Several influential studies also show that older adults are more likely to share and consume misinformation online, relative to younger users (Grinberg et al. 2019; Guess, Nagler, and Tucker 2019a; Allen et al. 2020). For instance, Guess, Nagler, and Tucker (2019a) find that individuals over the age of 65 shared false stories at more than twice the rate as the next youngest cohort. The authors stated that “no other demographic characteristic we examined—gender, income, education—had any consistent relationship with the likelihood of sharing fake news” (Guess, Nagler, and Tucker 2019b). Likewise, Alex Stamos, former chief security officer at Facebook, recently stated, “There’s good social science evidence that a lot of this is a boomer problem. Both on the left and the right, a lot of this stuff is being spread by folks who are our parents’ generation” (Kafka 2023).

Identifying differences across age cohorts is one thing, but correctly identifying the mechanisms at play is another. Scholars have forwarded a number of explanations, ranging from information environment factors to declining cognitive abilities associated with aging (Dodson, Powers, and Lytell 2015; Brashier and Schacter 2020).² Perhaps the most prominent account focuses on lower levels of digital literacy (Guess, Nagler, and Tucker 2019a; Moore and Hancock 2020, 2022). It is argued that older adults, as “nonnatives,” are simply less familiar with the digital media ecosystem (Hargittai and Dobransky 2017; Jo, Yang, and Yan 2022).

As digital literacy training is important in addressing misinformation online, it is natural to think that subpopulations especially vulnerable to this content would especially benefit from this type of intervention. Consequently, efforts have focused on training older citizens (Gringlas 2020; Span 2020) through the creation of training programs specifically tailored to educate older adults about risky information online. Moore and Hancock (2022), for instance, evaluate a novel digital literacy intervention for older adults motivated by the fact that “older adults are especially susceptible to fake news online, possibly because they are less digitally literate compared to younger individuals.” However, as noted, studies have found that older adults are no worse at identifying misleading content (Allcott and Gentzkow 2017; Brashier and Schacter 2020). Indeed, discernment abilities may *increase* with age (Badrinathan 2020; Calvillo et al. 2020; Peren Arin, Mazrekaj, and Thum 2023).

Resolving these conflicting findings is challenging. While it is relatively easy to establish a correlational link between age and engagement with dubious news, establishing a clear causal chain for *why* there are age differences is more difficult. Existing work posits both cohort and life-course explanations. Even focusing just on life-course explanations, there are many factors that change across the lifespan (e.g., income, education, life experience,

2. We address a cognitive account of age disparities in more detail in Lyons (2023).

personality (Boyce, Wood, and Powdthavee 2013)). Yet, we are aware of no existing study that has sought to arbitrate which factors are most central (but see an overview in Brashier and Schacter 2020).

In summary, numerous empirical studies (and our own results below) show that older adults are, indeed, significantly more likely to engage in low-quality news sites. But *why* this is the case remains unsettled. Moreover, one of the most widely mentioned explanations, poor digital literacy in older cohorts, is inconsistent with laboratory studies of discernment.

Age, Partisanship, and Engagement with Dubious Online Content

While digital literacy may be *one* determinant of the online behavior of older Americans, it is a mistake to completely put aside core concepts understood to shape political behavior. Previous research demonstrates that multiple factors change across the lifespan, several of which are well-established determinants of political activities. In particular, older Americans tend to be stronger partisans and exhibit more affective polarization, differences that predate the internet age.

To begin, due to increasing interest in news and politics (Glenn and Grimes 1968; Moretto et al. 2022), older Americans consume far more news overall. This greater political interest also coincides with increasingly established partisan identities and concurrent increases in affective polarization (Phillips 2022). Thus, older news consumers may exhibit greater directional motivations when encountering dubious news and may therefore share this content on social media regardless of perceived accuracy (Osmundsen et al. 2021; Pennycook and Rand 2021; Sirlin et al. 2021). Such directional motives may reflect social utility (e.g., sharing to strengthen relationships) (Duffy, Tandoc, and Ling 2020). As attitudes crystallize, partisan identity becomes more entrenched, and interest in politics increases across adulthood (Glenn and Grimes 1968; Converse 1969; Sears and Funk 1999; Hobbs 2019; Phillips 2022), derogation of the out-party (Osmundsen et al. 2021) may be an especially salient motive for older consumers. Consistent with this account is that polarization has increased the *most* among older Americans over the last three decades despite this demographic being the least likely to be engaged online (Boxell, Gentzkow, and Shapiro 2017).

Partisanship has long been understood as a foundational factor in determining political behaviors. Specifically, research consistently identifies partisanship or polarization as a driving force behind the consumption and spread of misinformation (Humprecht 2019; Pretus et al. 2023; Rao, Morstatter, and Lerman 2022). Indeed, some recent work argues that partisan polarization is *the* primary factor behind the sharing of political misinformation (Osmundsen et al. 2021).

It is worth noting nuance as well. If affective polarization increases with age, driven by increased interest in and consumption of political media (Garrett, Long, and Jeong 2019; Kim, Broussard, and Barnidge 2020; Suk et al. 2022), it may in turn then be associated with increases in both political knowledge (Kim, Broussard, and Barnidge 2020; Suk et al. 2022) and misperceptions (Garrett, Long, and Jeong 2019; Jenke 2024), which may be two sides of the same coin for the politically engaged (Allcott et al. 2020). In other words, affect-infused political interest can drive surveillance knowledge, but also can make individuals more vulnerable to (congenially) biased framing of real events. As a result, when considering evaluations of news headlines, we would expect that age could be linked to *recognition* of false content, but also *endorsement* of congenial hyperpartisan content.

Research Questions and Study Design

Is engagement with false news among older Americans mostly a function of greater partisanship that comes with age, or primarily the result of some other age-related factor like digital literacy? What empirical implications would help arbitrate between these competing accounts?

We begin by replicating the seemingly conflicting findings that older American adults are more likely to consume false news but are no worse at recognizing it. To our knowledge, trace data and discernment data have not previously been collected for the same sample. Instead, studies of online behavior typically rely on nonrepresentative (albeit large) administrative datasets (Grinberg et al. 2019). Discernment studies examining age differences (such as those reanalyzed in Brashier and Schacter 2020) have tended to rely on participants recruited from convenience samples such as Amazon's Mechanical Turk or SurveyMonkey (Allcott and Gentzkow 2017; Pennycook and Rand 2018). By replicating both within one dataset, we remove concerns that conflicting findings may merely be a statistical artifact of different sampling strategies.

Research Question 1 (news consumption):

(RQ1) Are older Americans more likely to engage with dubious news than younger cohorts?

Research Question 2 (news discernment):

(RQ2a) Are older Americans more or less capable than younger cohorts of discerning true and false news headlines?

(RQ2b) Do older Americans perceive false news headlines to be more or less accurate relative to younger cohorts?³

3. This specific test was preregistered in our initial study (<https://osf.io/94x5b>). The remaining analyses of age differences are exploratory.

Next, if the primary mechanism explaining older Americans' higher rates of engagement with false news is lower digital literacy, it seems reasonable to expect that digital literacy training should have a *larger* beneficial effect among respondents in this group.⁴ We explore whether a digital literacy intervention (Guess et al. 2020a)—among the most effective at reducing vulnerability to misinformation (Offer-Westort, Rosenzweig, and Athey 2024)—is more effective among older respondents.⁵

Research Question 3 (digital literacy intervention):

(RQ3) Are digital literacy interventions especially effective among older Americans?

The above research questions primarily aim to highlight the limitations—or at least incompleteness—of popular accounts of higher engagement among older Americans with false news. However, we also use our data to explore whether partisanship, which increases with age, can help account for this finding.

Research Question 4 (calcified partisanship):

(RQ4a) Do older Americans have stronger partisan attachments?

(RQ4b) Do older Americans exhibit higher levels of affective polarization?

(RQ4c) Do older Americans demonstrate higher levels of congeniality bias in evaluating news content?

Finally, if older Americans demonstrate higher levels of congeniality bias, it is reasonable that they may also be more likely to rate hyperpartisan news as accurate. Hyperpartisan domains can often be difficult edge cases of what constitutes “fake news.” Yet, as a borderline case, it is perhaps surprising that hyperpartisan content has not received more direct attention in survey tasks (particularly those examining age differences). Thus, higher susceptibility to hyperpartisan content generally rather than strictly false news content may provide a better explanation for apparently conflicting findings. This expectation is further supported by research on “clickbait” headlines that make appeals to partisan emotions. Like hyperpartisan news, partisan clickbait headlines can be viewed as low-quality (but not necessarily false) news content that is salient to those with strong partisan attachments. This is

4. A more traditional approach to arbitrating between competing claims might be to include some measure of digital literacy (Guess and Munger 2023) as a covariate. Unfortunately, our surveys were originally designed for a different question and did not include a measure of digital literacy (most of which were developed after our survey was fielded anyway). In addition, including both variables in the same specification would induce posttreatment bias if digital literacy is itself a function of age (Montgomery, Nyhan, and Torres 2018). Instead, we focus on where these competing theories generate different empirical expectations.

5. Our expectation—that training should not have larger effects across age, and thus not support a skills-based account—is consistent with recent experimental work showing that an intervention focused on helping respondents identify “troll” accounts worked no better among older adults (Lees et al. 2023).

relevant, because older Americans appear to exhibit a preference for such headlines (Munger et al. 2020), even when incentivized to identify correct information (Luca et al. 2022).

Research Question 5 (hyperpartisan engagement):

(RQ5a) Do older Americans perceive hyperpartisan news headlines to be more or less accurate relative to younger cohorts?

(RQ5b) Do older Americans demonstrate higher levels of congeniality bias for hyperpartisan news content?

To test these expectations, we conduct a secondary analysis of three separate surveys collected before and after the 2018 US elections. There are three features of this dataset that make it especially suitable for our purposes. First, the dataset includes both surveys and digital trace data, increasing our confidence that any difference between exposure and discernment is not an artifact of different sampling strategies. Second, one survey included random assignment to a digital literacy intervention, where half of respondents in the third survey were exposed to a set of ten brief “Tips to Spot False News.” Importantly, this intervention improved discernment overall. Therefore, we are able to test whether older respondents were especially responsive (suggesting an initial literacy deficit). Third, our data includes measures of strength of partisanship and affective polarization as well as evaluations of hyperpartisan headlines (although it does include a measure of digital literacy itself).

We summarize our research questions and preview our findings in [table 1](#).

Methods

Survey Data Collection

We draw on data from three two-wave survey panels⁶ conducted by the survey company YouGov during and after the 2018 US midterm elections.⁷ By collecting data across three surveys, we can replicate our analyses across time, samples, and stimuli. Our data come from: a survey fielded June 25–July 3, 2018 (wave 1; N = 1,718) and July 9–17, 2018 (wave 2; N = 1,499); a survey fielded October 19–26 (wave 1; N = 3,378) and October 30–November 6, 2018 (wave 2; N = 2,948); and a survey fielded November 20–December 27, 2018 (wave 1; N = 4,907) and December 14, 2018–January 3, 2019 (wave 2; N = 4,283). Respondents were selected by YouGov’s matching and weighting algorithm to approximate the demographic and political attributes of the US population

6. As mentioned, we make use of existing data that was collected for other studies. In our analyses, the two-wave panel structure only comes into play when examining decay of the literacy training effect in the Nov./Dec. study.

7. These surveys included multiple orthogonal experiments (Guess et al. 2020a, 2020b; Berlinski et al. 2023).

Table 1. Summary of research questions and findings.

Question	Finding	Results
<i>News consumption:</i> Are older Americans more likely to engage with dubious news?	Yes	Figure 1/ Table 2
<i>News discernment:</i> Are older Americans more or less capable than younger cohorts of discerning true and false news headlines? Do older Americans perceive false news headlines to be more or less accurate relative to younger cohorts?	Not distinguishable Less accurate	Table 3 Table 3
<i>Digital literacy intervention:</i> Are digital literacy interventions especially effective among older Americans?	Not distinguishable	Table 4
<i>Partisanship:</i> Do older Americans have stronger partisan attachments? Do older Americans exhibit higher levels of affective polarization? Do older Americans demonstrate higher levels of congeniality bias in evaluating news content?	Yes Yes Yes	Figure 3 Figure 3 Figure 4/ Table 5
<i>Hyperpartisan content:</i> Do older Americans perceive hyperpartisan news headlines to be more or less accurate relative to younger cohorts? Do older Americans demonstrate higher levels of congeniality bias for hyperpartisan news content?	More accurate Yes	Table 6 Table 7

(see [Supplementary Material Appendix A](#)). Participants were ineligible to take part in more than one study.

Measures

Headline Rating

In each survey, we asked respondents to evaluate the accuracy of a number of headlines on a four-point scale. The articles, all of which appeared “in the wild” during the 2018 midterms, were published by actual sources and were balanced within each group in terms of their partisan congeniality. In total, we selected four mainstream news articles that were congenial to Democrats and four that were congenial to Republicans, and two pro-Democrat and two pro-Republican false news articles. Respondents also rated the accuracy of four hyperpartisan news headlines. These headlines were chosen from outlets defined as hyperpartisan by [Pennycook and Rand \(2019\)](#). To keep distinction from false headlines, we selected hyperpartisan headlines that had not been fact-checked as false. All headlines are presented in [Supplementary Material Appendix B](#).

In the first wave of each survey, respondents evaluated a random subset of headlines from each slant-and-veracity subcategory (e.g., pro-Republican false news, etc.) on a scale ranging from “Not at all accurate” (1) to “Very accurate” (4). In the second wave, respondents evaluated all sixteen headlines. The order of the headlines was randomized within wave for each respondent.

We look at perceived accuracy for each news type (mainstream, false, and hyperpartisan). For these, we also include a measure of headline congeniality. Congeniality is coded at the headline level for partisans to indicate that a story is consistent with the respondents' partisan leanings (e.g., a Democrat evaluating a story that is favorable to a Democrat) and is zero otherwise. We also examine the ability to distinguish mainstream from false news—*discernment*—as $\text{mean}(\text{mainstream news accuracy}) - \text{mean}(\text{false news accuracy})$. For comparison, we also compute a discernment measure using hyperpartisan news using $\text{mean}(\text{mainstream news accuracy}) - \text{mean}(\text{hyperpartisan news accuracy})$. Descriptive statistics are listed in [Supplementary Material table A.1](#).

Media Exposure Data/Trace Data

We also analyze media exposure using behavioral data of respondents' web visits, collected unobtrusively. Data is available from users' laptop or desktop computers as well as their mobile devices and tablets. Our measures come from a period immediately following each survey.⁸

8. See [Supplementary Material Appendix A](#) for details, including lists of domains used.

We first created a binary measure of whether respondents made one or more visits to dubious news sites, as well as a binary measure for mainstream news sites. We also look at a count of total visits to either set of news domains. However, it is worth noting that the distribution was highly skewed; 90–94 percent of respondents visited zero dubious news sites, and the distribution among nonzero respondents had a long right tail (June/July $M = 1.75$, $SD = 18.26$, $\min = 0$, $\max = 438$; Oct./Nov. $M = 0.43$, $SD = 3.24$, $\min = 0$, $\max = 75$; Nov./Dec. $M = 0.21$, $SD = 1.37$, $\min = 0$, $\max = 25$). Our binary measure of *dubious news exposure* is coded as 1 if the respondent visited any of the domains in our list (June/July: 10 percent, Oct./Nov.: 7 percent, Nov./Dec.: 6 percent) and 0 otherwise. We also created a binary measure of *mainstream news exposure* (June/July: 73 percent, Oct./Nov.: 60 percent, Nov./Dec.: 52 percent).

Importantly, not all respondents who were part of our survey opted into the web-tracking portion of our study. Thus, our sample size using this data decreases (June/July web tracking $n = 1,312$ (76 percent), Oct./Nov. $n = 2,103$ (63 percent), Nov./Dec. $n = 1,069$ (22 percent), total web tracking $n = 4,484$ (50 percent)).⁹

Demographics and Sociopolitical Measures

We also measure objective political knowledge, political interest, dichotomous indicators of Democrat and Republican affiliation (including leaners), college education, gender, and nonwhite racial background. We measure affective polarization as the difference in people's feelings toward their preferred party and the opposition party on 101-point feeling thermometers (including leaners but excluding independents). (The full question list is in [Supplementary Material Appendix C.](#))

Analytic Approach

We use OLS regression models with age as our predictor, and include party, political knowledge, political interest, college education, gender, and nonwhite racial background as covariates unless otherwise noted. Except in the case of the experimental results for RQ3, these models reflect the application of survey weights. In all analyses, we pool across surveys.

Our primary analyses use dichotomous indicators for a standard set of age groups, as fitting linear interaction models can mask nonlinearities (Hainmueller, Mummolo, and Xu 2019). Further, studies employing trace data summarized above suggest that it is the oldest subgroup of users, in particular, who are most likely to consume dubious news, specifically those 60+ (Guess, Nyhan, and

9. The decline between surveys reflects the lack of available respondents who (a) participated in the web tracking and (b) did not participate in our earlier waves of data collection.

Reifler 2020; Moore, Dahlke, and Hancock 2023). As such, we look at the size of prior exposure effects across the same age categories as used in these studies. However, for all results, we provide the linear interaction models in supplementary analyses shown throughout [Supplementary Material Appendix D](#).

For headline accuracy rating models, we include fixed effects for each headline to account for differing baseline levels of plausibility and cluster at the respondent level to account for correlations between their ratings across headlines. The headline models also include a measure of headline congeniality computed based on a given headline's slant and the respondent's party identity. We also include covariates and use survey weights.

Results

Older Americans Visit More Dubious and More Mainstream News Sites (RQ1)

We first look at mainstream and dubious news visits (binary and count, pooled across surveys). We find a relationship between age and both dubious and mainstream news exposure (binary measures) as well as the volume of consumption for each (count measure). For instance, respondents over 60 made about 0.5 ($p = 0.018$) more visits to dubious domains during our data collection period than did those between 18 and 29 and made about 14 more mainstream news domain visits than did the youngest cohort ($p = 0.009$). Although these relationships are significant for a linear age term ([Supplementary Material table D.2](#)), we find that the "effect" of age is concentrated among those over 60 years old ([table 2](#) and [figure 1](#)). Finally, we note that the relationship between age and news consumption is stronger prior to accounting for political interest and knowledge ([Supplementary Material table D.3](#)). Older Americans are more politically interested and knowledgeable, and those higher in these traits tend to consume more news (of both dubious and mainstream varieties). In sum, though older subgroups are more often exposed to dubious news domains, this may in part be a symptom of their greater interest in news and politics. However, age appears to matter after accounting for this trait. Critically, we also find that dubious news visits as a *proportion* of all news visits still increase with age ([Supplementary Material table D.5](#)), so it is unlikely that this pattern is explained just by higher engagement with news generally.

Older Americans are Generally No Worse at Evaluating News Accuracy (RQ2)

We test headline evaluation using multiple model specifications ([table 3](#)). We look at the perceived accuracy of both mainstream and false news as well as the discernment between the two as our outcomes.

Table 2. Association of age with news visits, OLS regression estimates.

	Dubious binary	Dubious count	Mainstream binary	Mainstream count
Age 30–44	0.012 (0.013) [0.37]	–0.054 (0.121) [0.654]	–0.062 (0.036) [0.08]	–0.642 (4.053) [0.874]
Age 45–59	0.021 (0.012) [0.064]	0.413 (0.287) [0.149]	0.019 (0.035) [0.581]	3.794 (4.441) [0.393]
Age 60+	0.049 (0.013) [0.000]	0.547 (0.232) [0.018]	0.060 (0.033) [0.075]	14.361 (5.485) [0.009]
Control variables	✓	✓	✓	✓
R ²	0.08	0.01	0.06	0.02
N	4,365	4,365	4,365	4,365

Note: Robust SEs are in parentheses and two-sided *p*-values in brackets. Binary exposure is coded as 1 if the respondent visited any such domain and 0 otherwise. Age 18–29 is the reference group. All models include controls for Democrat, Republican, college education, political knowledge, political interest, gender, and nonwhite racial background.

Our results show that older Americans perceive both false and mainstream headlines as less accurate, resulting in discernment scores that are generally no worse than younger cohorts. One interpretation is that there is not a media literacy deficit among older news consumers. To address this more formally, we consider whether digital literacy training has differential effects across age.

No Differential Effects of a Digital Literacy Intervention (RQ3)

We randomly assigned respondents with probability 0.5 to the “Tips for spotting false news” stimuli in wave 1 of the November/December study (figure 2).¹⁰ This intervention took place before the headline rating task.

Results in table 4 show that literacy training had no differential effect across respondent age categories. The results show that there is some evidence that the tips worked better for adults aged 45–59 ($\beta = -0.15$, $p = .031$) in identifying false news. However, this does not hold for respondents

10. Due to a programming error, all respondents in the June/July and October/November studies were exposed to the tips intervention. To make sure this does not affect our conclusions about the association of age with perceived accuracy and discernment, we present results using the control group from the Nov./Dec. survey in [Supplementary Material table D.6](#).

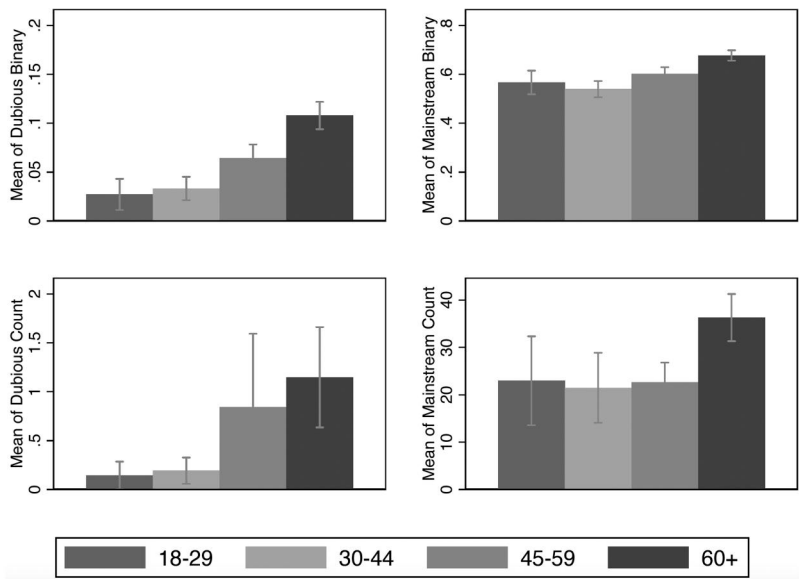


Figure 1. Age and news visits (by age groups). Figure shows descriptive means. The binary measure of dubious (mainstream) news exposure is coded as 1 if the respondent visited any such domain and 0 otherwise. Error bars are 95 percent confidence intervals. Full results of models including a standard set of covariates are shown in [table 2](#). The vertical axes are not equivalent in order to better show age differences.

aged 60+ as we would expect, nor does it replicate with a linear measure of age ([Supplementary Material table D.8](#)). Further, in [Supplementary Material table D.8](#), we examine results for the second wave and also find no reliable evidence that the treatment effect decayed faster for older cohorts. While failing to reject the null is not direct evidence of no effect heterogeneity by age, these results at least fail to support the theory that digital literacy is the prime driver of engagement with dubious news.

Explaining the Disagreement

Older Americans are More Subject to Partisan Bias in News Evaluation (RQ4)

A key way that older news consumers may differ is in their partisanship. Indeed, in our data, respondents aged 60+ were significantly more likely to strongly identify with a political party, and affective polarization consistently increases with age ([figure 3](#)), even after accounting for demographics

Table 3. Association of age with perceived accuracy of false and mainstream stories and with discernment, OLS regression estimates.

	Wave 1			Wave 2		
	False	Mainstream	Discernment	False	Mainstream	Discernment
Age 30–44	–0.035 (0.031) [0.264]	0.007 (0.024) [0.783]	0.041 (0.034) [0.221]	–0.035 (0.029) [0.233]	–0.023 (0.025) [0.365]	0.013 (0.027) [0.644]
Age 45–59	–0.150 (0.031) [0.000]	–0.102 (0.025) [0.000]	0.045 (0.036) [0.201]	–0.073 (0.029) [0.012]	–0.074 (0.025) [0.003]	0.000 (0.028) [0.988]
Age 60+	–0.160 (0.029) [0.000]	–0.160 (0.023) [0.000]	–0.003 (0.033) [0.927]	–0.051 (0.027) [0.061]	–0.059 (0.022) [0.008]	–0.008 (0.027) [0.765]
Control variables	✓	✓	✓	✓	✓	✓
Headline fixed effects	✓	✓		✓	✓	
R ²	0.15	0.17	0.17	0.16	0.17	0.24
N	19,886	39,765	9,944	34,718	69,433	8,682

Note: Robust SEs are in parentheses and two-sided *p*-values in brackets. Age 18–29 is the reference group. All models include controls for Democrat, Republican, college education, political knowledge, political interest, gender, and nonwhite racial background. Headline-level analyses also control for congeniality.

and political sophistication (see [Supplementary Material table D.10](#)). Respondents aged 60+ score about 11 points higher in affective polarization than do those aged 18–29 after accounting for covariates.¹¹

Next, we test whether older Americans are also more subject to partisan congeniality effects when evaluating headlines. We look at perceived headline accuracy as our outcome variable. As seen in [Supplementary Material table D.12](#) and [figure 4](#), older Americans, including those aged 45–59, but especially those over 60, are especially subject to partisan congeniality bias when evaluating headlines. This appears to hold across headline veracity ([table 5](#)). Finally, in [Supplementary Material tables D.15](#) and [D.16](#), we also partially replicate these patterns using beliefs in misperceptions about recent current events.

11. Note also that political interest and knowledge increase with age in our data ([Supplementary Material table D.11](#)).

"Tips to Spot False News"

Be skeptical of headlines. False news stories often have catchy headlines in all caps with exclamation points. If shocking claims in the headline sound unbelievable, they probably are.

Look closely at the URL. A phony or look-alike URL may be a warning sign of false news. Many false news sites mimic authentic news sources by making small changes to the URL. You can go to the site to compare the URL to established sources.

Investigate the source. Ensure that the story is written by a source that you trust with a reputation for accuracy. If the story comes from an unfamiliar organization, check their "About" section to learn more.

Watch for unusual formatting. Many false news sites have misspellings or awkward layouts. Read carefully if you see these signs.

Consider the photos. False news stories often contain manipulated images or videos. Sometimes the photo may be authentic, but taken out of context. You can search for the photo or image to verify where it came from.

Inspect the dates. False news stories may contain timelines that make no sense, or event dates that have been altered.

Check the evidence. Check the author's sources to confirm that they are accurate. Lack of evidence or reliance on unnamed experts may indicate a false news story.

Look at other reports. If no other news source is reporting the same story, it may indicate that the story is false. If the story is reported by multiple sources you trust, it's more likely to be true.

Is the story a joke? Sometimes false news stories can be hard to distinguish from humor or satire. Check whether the source is known for parody, and whether the story's details and tone suggest it may be just for fun.

Some stories are intentionally false. Think critically about the stories you read, and only share news that you know to be credible.

[These tips are taken verbatim from the original tips published by Facebook.]

Figure 2. Media literacy intervention.

Older Americans are More Credulous to Hyperpartisan News in Surveys, but Prior Studies Elided the Category (RQ5)

Though related to the partisan motivation account touched on above, the next explanation for the disjuncture between survey and trace data focuses on methodology. When examining sharing patterns or exposure through web tracking, researchers often use domain lists that include websites that are frequently flagged for publishing flawed and misleading news stories (Grinberg et al. 2019; Guess, Nyhan, and Reifler 2020). However, coding these websites at the domain level reveals sharing of or exposure to the broader category of low-quality news, and not exclusively *false* news (see for instance the “red” and “orange” lists—which account for most shares—vs. “black” lists in these studies). In essence, then, these “false news sites,” as we too have called them in our prior work—may often be more accurately classified as hyperpartisan news sites (in Supplementary Material table D.1, we include examples of prior studies that classify the sources for our hyperpartisan headlines as “fake” or “false” news sites).

However, when older Americans’ ability to discern news has been judged, the surveys have typically included stimuli from (accurate) mainstream headlines contrasted with headlines that had been demonstrably fact-checked as false (Allcott and Gentzkow 2017; Pennycook and Rand 2018). Given the potential epistemological rabbit hole of defining what is truth, relying on fact-checked claims when studying misinformation and false news is a reasonable approach. However, focusing on demonstrably false content elides

Table 4. Interaction effects of age and media literacy training treatment on discernment, OLS regression estimates.

	False	Mainstream	Discernment	False	Mainstream	Discernment
News tips intervention	−0.196 (0.020) [0.000]	−0.046 (0.017) [0.006]	0.127 (0.027) [0.000]	−0.126 (0.058) [0.030]	−0.038 (0.046) [0.401]	0.066 (0.067) [0.324]
Age 30–44				−0.043 (0.050) [0.396]	−0.006 (0.041) [0.887]	0.034 (0.050) [0.499]
Age 45–59				−0.089 (0.049) [0.070]	−0.097 (0.040) [0.015]	−0.026 (0.051) [0.618]
Age 60+				−0.195 (0.050) [0.000]	−0.131 (0.039) [0.001]	0.050 (0.052) [0.339]
Age 30–44 x news tips				−0.052 (0.072) [0.473]	0.024 (0.058) [0.672]	0.086 (0.080) [0.282]
Age 45–59 x news tips				−0.153 (0.071) [0.031]	−0.033 (0.057) [0.555]	0.148 (0.081) [0.068]
Age 60+ x news tips				−0.049 (0.069) [0.484]	−0.066 (0.054) [0.216]	0.006 (0.079) [0.935]
Control variables				✓	✓	✓
Headline fixed effects	✓	✓		✓	✓	
R ²	0.04	0.07	0.01	0.18	0.17	0.19
N	9,813	19,623	4,907	9,797	19,591	4,899

Note: Robust SEs in parentheses and two-sided *p*-values in brackets. Models testing for age interactions include controls for Democrat, Republican, college education, political knowledge, political interest, gender, and nonwhite racial background, and headline congeniality (in headline level analyses).

hyperpartisan content—emotional, slanted, and often misleading, but based on actual events and not necessarily categorically false. We rectify this by including both false and hyperpartisan headlines in our accuracy rating task.

For these analyses, we follow the same model specifications as our prior models addressing perceived accuracy of mainstream and false news and

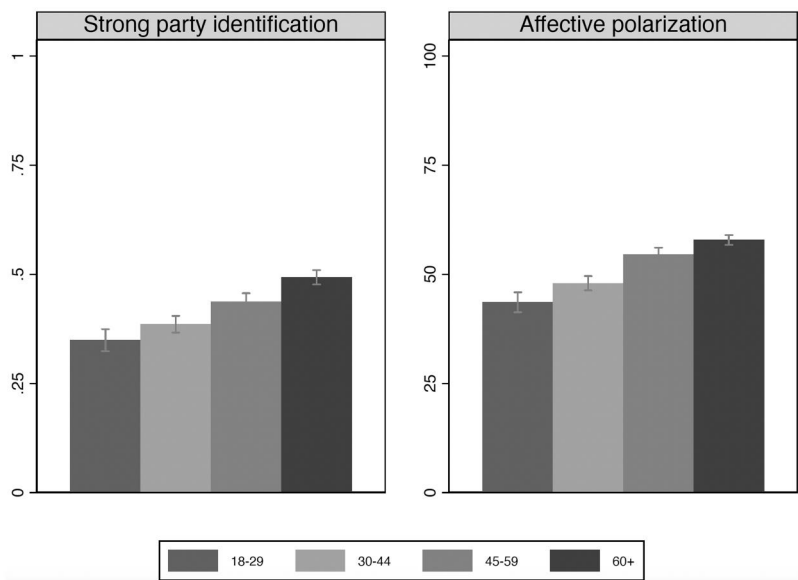


Figure 3. Party attachment and affective polarization by age group. Figure shows descriptive means. For strong party attachment, independents are coded as 0; for affective polarization, independents are by definition dropped. Error bars are 95 percent confidence intervals. Full results of models including a standard set of covariates are shown in [Supplementary Material table D.10](#).

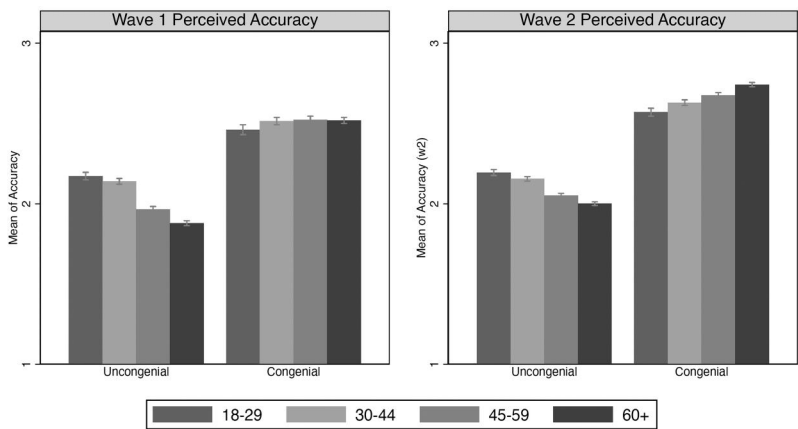


Figure 4. Age and congeniality effects (pooled news types). Figure shows descriptive means. Error bars are 95 percent confidence intervals. Full modeled results are shown in [Supplementary Material table D.12](#).

Table 5. Interaction effects of age and message congeniality on perceived accuracy by news type, OLS regression estimates.

	Wave 1		Wave 2	
	False	Mainstream	False	Mainstream
Age 30–44	–0.046 (0.037) [0.210]	–0.027 (0.030) [0.368]	–0.054 (0.035) [0.118]	–0.044 (0.030) [0.145]
Age 45–59	–0.171 (0.036) [0.000]	–0.207 (0.030) [0.000]	–0.111 (0.034) [0.001]	–0.153 (0.030) [0.000]
Age 60+	–0.224 (0.034) [0.000]	–0.275 (0.029) [0.000]	–0.135 (0.031) [0.000]	–0.169 (0.026) [0.000]
Congenial	0.475 (0.042) [0.000]	0.363 (0.032) [0.000]	0.534 (0.036) [0.000]	0.475 (0.031) [0.000]
Age 30–44 x Congenial	0.031 (0.052) [0.543]	0.091 (0.041) [0.028]	0.05 (0.046) [0.279]	0.057 (0.040) [0.151]
Age 45–59 x Congenial	0.055 (0.051) [0.286]	0.264 (0.041) [0.000]	0.097 (0.046) [0.033]	0.201 (0.039) [0.000]
Age 60+ x Congenial	0.151 (0.048) [0.002]	0.28 (0.039) [0.000]	0.203 (0.042) [0.000]	0.264 (0.035) [0.000]
Control variables	✓	✓	✓	✓
Headline fixed effects	✓	✓	✓	✓
R ²	0.15	0.18	0.16	0.17
N	19,886	39,765	34,718	69,433

Note: Robust SEs are in parentheses and two-sided *p*-values in brackets. Age 18–29 is the reference group. All models include controls for Democrat, Republican, college education, political knowledge, political interest, gender, and nonwhite racial background.

discernment between them but use perceived accuracy of hyperpartisan news and mainstream-hyperpartisan discernment as outcomes. While older Americans were more likely than younger cohorts to rate both false and mainstream headlines as inaccurate (table 3), there was a departure when

Table 6. Association of age with perceived accuracy of hyperpartisan news and mainstream-hyperpartisan discernment, OLS regression estimates.

	Accuracy of hyperpartisan news		Mainstream-hyperpartisan discernment	
	Wave 1	Wave 2	Wave 1	Wave 2
Age 30–44	0.048 (0.030) [0.116]	0.051 (0.030) [0.088]	–0.036 (0.032) [0.250]	–0.074 (0.028) [0.008]
Age 45–59	0.012 (0.030) [0.682]	0.061 (0.029) [0.039]	–0.112 (0.033) [0.001]	–0.134 (0.028) [0.000]
Age 60+	0.011 (0.028) [0.706]	0.107 (0.028) [0.000]	–0.169 (0.031) [0.000]	–0.166 (0.028) [0.000]
Control variables	✓	✓	✓	✓
Headline fixed effects	✓	✓		
R ²	0.17	0.19	0.13	0.19
N	19,883	34,723	9,944	8,683

Note: Robust SEs are in parentheses and two-sided *p*-values in brackets. Age 18–29 is the reference group. All models include controls for Democrat, Republican, college education, political knowledge, political interest, gender, and nonwhite racial background, as well as headline congeniality.

rating hyperpartisan headlines, which they saw as more accurate in some cases (i.e., in wave 2) (table 6). Notably, older respondents exhibit stronger congeniality bias for hyperpartisan headlines than other news types (table 7).

This is also reflected in the second discernment score, which subtracted hyperpartisan accuracy from mainstream accuracy. While older Americans were generally no worse at discerning between mainstream and false headlines, they were consistently worse at discerning between mainstream and hyperpartisan headlines (see [Supplementary Material figure D.1](#)). In this way, the ostensible gap between trace data and survey data can be partially resolved. Further, this methodological explanation can be fit within the theoretical account regarding increasing partisan animosity among older news consumers and preference for derogative content ([Osmundsen et al. 2021](#)).

Table 7. Interaction effects of age and message congeniality on perceived accuracy of hyperpartisan news, OLS regression estimates.

	Accuracy of hyperpartisan news	
	Wave 1	Wave 2
Age 30–44	–0.002 (0.035) [0.951]	–0.010 (0.035) [0.763]
Age 45–59	–0.125 (0.034) [0.000]	–0.034 (0.035) [0.320]
Age 60+	–0.147 (0.033) [0.000]	–0.052 (0.031) [0.098]
Congenial	0.481 (0.043) [0.000]	0.640 (0.040) [0.000]
Age 30–44 x Congenial	0.133 (0.053) [0.011]	0.161 (0.050) [0.001]
Age 45–59 x Congenial	0.346 (0.053) [0.000]	0.244 (0.050) [0.000]
Age 60+ x Congenial	0.383 (0.051) [0.000]	0.384 (0.045) [0.000]
Control variables	✓	✓
Headline fixed effects	✓	✓
R ²	0.17	0.19
N	19,883	34,723

Note: Robust SEs are in parentheses and two-sided *p*-values in brackets. Age 18–29 is the reference group. All models include controls for Democrat, Republican, college education, political knowledge, political interest, gender, and nonwhite racial background.

Discussion

Older news consumers are often identified as among the most vulnerable to dubious news online, with blame often attributed to generational digital illiteracy. However, little empirical work has demonstrated the source of this age group’s greater engagement. In this study, we replicate past work that shows older Americans are no worse at discerning true from false news in

surveys, despite their greater exposure to dubious websites as measured in digital trace data. We likewise show that digital literacy training is no more or less effective among older respondents. Accordingly, we argue that digital literacy deficits are likely not the sole culprit, although they may still play a role. Instead, we emphasize motivational and methodological explanations centered on increasing entrenched partisan identity—closely associated with greater political interest and media consumption—across the lifespan (Glenn and Grimes 1968; Sears and Funk 1999; Plutzer 2002; Phillips 2022).

First, we emphasize that some of the differences in exposure across age groups may be attributable to the sheer volume of news that older Americans consume. Likely reflecting generally high political interest (Glenn and Grimes 1968; Moretto et al. 2022), we find that older cohorts consume more dubious news as well as mainstream news.

Second, perceived accuracy and engagement decisions are not necessarily related (Pennycook and Rand 2021; Sirlin et al. 2021), a disjuncture that seems to be more apparent among older Americans (Munyaka, Hargittai, and Redmiles 2022). This suggests that other motivations beside accuracy could be driving engagement choices. We show that older Americans consistently display more congeniality bias when evaluating headlines. One implication of this bias is that older news consumers could be more likely to visit or share news that derogates the other party in spite of its inaccuracy (Osmundsen et al. 2021). A weakness for emotional, out-group derogating news—but not fully fabricated news—is also reflected in older Americans' more credulous rating of hyperpartisan headlines, but not false headlines. Although prior work that suggests older Americans are better at discerning between higher- and lower-quality news did not account for this content category, it may better reflect the sort of news often classified as “fake” given the domain lists included in such studies.

In this way our account is closely linked to existing research on the role of affective polarization in engaging with dubious news content. Previous work finds that affective polarization, fueled by heightened engagement with political media (Garrett, Long, and Jeong 2019; Kim, Broussard, and Barnidge 2020; Suk et al. 2022), is linked to higher levels of both political knowledge and misperceptions (Garrett, Long, and Jeong 2019; Allcott et al. 2020; Kim, Broussard, and Barnidge 2020; Suk et al. 2022; Jenke 2024). Essentially, when people's emotions become intertwined with their political interests, they may become more attentive to political information, but also more susceptible to biased interpretations of factual events. When turning to the role of age in news evaluations, we similarly find that older news consumers, who also happen to be more engaged and emotionally invested in politics, are more likely to recognize false content but also to endorse slanted hyperpartisan content.

These findings have several important implications for those interested in reducing engagement with dubious news among older cohorts. First, our findings imply that while digital literacy training may be useful in general (Guess et al. 2020a), older Americans do not suffer a particular deficit and interventions aiming to improve the quality of online news sharing may be better served by targeting the increasing partisan animosity among these news consumers. Depolarization techniques (Voelkel et al. 2023) may be tailored for these partisans more specifically (but see Broockman, Kalla, and Westwood 2023). Further on the topic of interventions, because older adults tend to perceive both false and mainstream news as less accurate than other age groups do (Munyaka, Hargittai, and Redmiles 2022), interventions that target increased trust in legitimate sources may be especially important, given that mainstream news constitutes the vast majority of all news consumed (Acerbi, Altay, and Mercier 2022).

Additionally, our results invite comparative research on vulnerability to misinformation across the lifespan; age differences in engagement may be less pronounced in less polarized contexts (Humprecht 2019). By the same token, in terms of theoretical implications, these results further suggest that misinformation researchers should consider social contexts that often underlie behavior alongside universal cognitive explanations (Adam-Troian et al. 2021).

Finally, these results have methodological implications. Researchers should not only think carefully about outcomes of interest—as belief and engagement behaviors frequently diverge—but also about how our approach to measurement can stretch or distort the concept we are studying in subtle ways. Our results indicate that part of the source for the conflict between findings based on experimental and digital trace data may be how the outcome variable was operationalized. While the former has typically focused on news content that has been fact-checked as false, the latter regularly includes slanted (but not explicitly false) hyperpartisan content that is intermixed with explicitly false content on the same domains. When we instead conduct experiments using hyperpartisan news, we do indeed find that older cohorts are more susceptible to low-quality news content—especially when congenial. This implies that scholars may wish to more regularly include hyperpartisan news in their research designs. But it also indicates that the underlying problem facing older cohorts may not actually be engagement with misinformation *per se*.

Of course, these findings should also be considered in light of their limitations. To begin, this is an exploratory (not preregistered) analysis based on a secondary analysis of a previously completed study. In addition, there is always some level of error associated with public opinion research, and we rely exclusively on nonprobability samples. Further, we cannot test all possible contributors to age differences in consumption and sharing, such as shifts

in interpersonal trust or sets of social ties (Brashier and Schacter 2020; Guess et al. 2021). Similarly, we focused on the US context, in light of prior work revealing older American news users as especially vulnerable as a group. However, there may be cross-national differences based around differing political and media systems (Humprecht, Esser, and Van Aelst 2020) (but see Arechar et al. 2023). In terms of measurement, our web-tracking data is limited in that it captures only visits and cannot assess exposure via social news feeds, and likewise does not account for offline exposure, which is still common among older users. Our analysis of congeniality bias also cannot directly speak to the out-group derogating motives that prior work suggests drive engagement with dubious news. Instead, we must indirectly speculate that as such bias increases, so do such motives (Combs et al. 2009; Lelkes and Westwood 2017).

Finally, our results do not distinguish between age, period, and cohort (APC) effects. On this issue, it is worth considering Phillips's (2022) estimates of APC effects on affective polarization and partisan strength—our proposed drivers of news engagement. Affective polarization may increase both with age (due to age-related increases in partisan strength) and over time in the United States. Though the analysis found no clear cohort effects, “there is some evidence that the Baby Boomer generation is chronically lower in both in-party and out-party warmth” (p. 1484). Researchers might consider whether habits those currently over 60 picked up early in life have something to do with their current news use. Spending formative years under a system featuring just a few, largely reliable news outlets may have resulted in overly credulous news engagement, for instance. This is complicated by our finding that older adults find both false and mainstream news headlines to be less accurate than younger respondents do; it is only hyperpartisan news that older respondents especially endorse. However, this outcome could be understood in light of this cohort's chronically low warmth toward both parties.

Overall, our results challenge conventional wisdom regarding older news consumers using three large, nationally representative surveys that match survey and tracking data. Rather than victims of digital illiteracy, we find that older Americans typically recognize and disbelieve false news. However, they also demonstrate heightened partisan responses to news, and as such are particularly likely to engage with congenial hyperpartisan news content, much of which happens to be classified as false news in prior research based on domain-level classifications.

Supplementary Material

Supplementary Material may be found in the online version of this article: <https://doi.org/10.1093/poq/nfae044>.

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Data Availability

Replication data and documentation are available at <https://osf.io/zax5k/>.

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